

PROJECT
Econometrics

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PROJECT

The effect of the Veligo e-bike rental program on cycling intensity in the Paris region

Research question and motivation

I want to estimate the causal impact of the Veligo long-term e-bike rental program on cycling behavior in the Ile-de-France region. Veligo aims to reduce car usage and increase active mobility by offering subsidized long-duration access to e-bikes. The main outcome is cycling intensity (trips and distance), and secondary outcomes include substitution away from public transport or cars, and persistence after the rental period ends. Understanding whether the program meaningfully increases cycling (and for whom) is important for transport policy design and subsidy targeting.

Data needed:

1. Cycling outcomes
 - Trip counts and trip distance by area and time (weekly or daily).
 - If possible: station-level or municipality-level measures of bike trips.
2. Treatment intensity
 - Veligo uptake: number of active rentals/subscriptions by municipality (or by postcode) and by week/month.
 - Program rollout dates, eligibility rules, and any changes in pricing/subsidy over time.
3. Control variables
 - Weather (temperature, rainfall, wind) by week and area.
 - Public transit disruptions (strikes, line closures) and major events.
 - Local socioeconomic controls (population, income, commuting patterns) at municipality level.

How I will collect the data:

1. Open data for biking outcomes
 - Velib Metropole open data: trip-level or aggregated station-level ride counts (depending on availability), with timestamps and station locations.
 - Paris/ Ile-de-France open data portals for cycling counters (if available), which provide continuous counts of cyclists at fixed locations.
2. Veligo program data
 - Ile-de-France Mobilites / Veligo public statistics: monthly or quarterly uptake, potentially by department or municipality. If municipality-level uptake is not public, I would request aggregated data from the operator or use department-level uptake as the treatment intensity.

3. Weather and events
 - Meteo France or publicly available historical weather datasets (daily/weekly).
 - Public transit strike calendars and disruption reports from official transit operators (RATP/SNCF) where possible.

Empirical strategy to establish causality

Main method: Difference in Differences (DiD) with staggered adoption / intensity

Idea

Uptake and program intensity vary across space and time. I will compare changes in cycling outcomes in areas with higher Veligo uptake to changes in areas with lower uptake, before and after the program expansion periods, controlling for time effects and area fixed characteristics.

Baseline regression (panel DiD):

$$Y_{it} = \beta_0 + \beta_1 * \text{VeligoIntensity}_{it} + \text{areaFE}_i + \text{timeFE}_t + X_{it} * \gamma + \text{error}_{it}$$

Where:

- Y_{it} is cycling intensity in area i at time t (weekly trips, distance, or counter counts).
- $\text{VeligoIntensity}_{it}$ is the number of active rentals per capita in area i at time t (or an indicator for high-uptake areas post-expansion).
- areaFE_i are area fixed effects that absorb time-invariant differences between areas (cycling culture, infrastructure baseline).
- timeFE_t are week fixed effects that absorb common shocks (seasonality, fuel prices, national trends).
- X_{it} includes weather and strike controls.

Coefficient of interest:

β_1 measures how cycling changes when Veligo intensity increases, net of area and time fixed effects and controls.

Key identifying assumption

Parallel trends: In the absence of the program (or in the absence of increases in uptake), high-uptake and low-uptake areas would have followed similar time trends in cycling outcomes after controlling for fixed effects and observables like weather.

Validation and robustness

1. Event-study version of DiD
 - I will estimate dynamic effects around program changes to check for pre-trends (effects before uptake increases should be near zero).
2. Placebo tests
 - Use outcomes that should not respond strongly (for example, cycling counters far from residential zones if the treatment is residential uptake, depending on geography).

- Use placebo adoption dates to verify no spurious effects.
3. Alternative control groups
 - Compare to areas with similar baseline cycling and demographics (matching on pre-period levels).
 4. Sensitivity to weather and strikes
 - Include flexible weather controls and strike indicators, and test whether results change materially.

Expected contribution

This project will provide evidence on whether subsidized long-term e-bike rentals causally increase cycling activity, and whether effects are short-lived or persistent. The results can inform whether expanding Veligo-like programs is cost-effective compared to other mobility interventions such as infrastructure investments or purchase subsidies.